Pranjal Kalekar

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SUMMARY

Applied Data Scientist with strong expertise in designing and deploying scalable machine learning models using Python and contemporary toolkits. Collaborative problem solver adept at driving data-driven solutions from experimentation to production.

TECHNICAL SKILLS

- **Tech Languages:** Python, MySQL, MS SQL, R, MATLAB, No-SQL, C++, HTML/CSS
- **Databases:** MongoDB, MariaDB, Neo4j, Apache Spark, PostgreSQL, NoSQL, RDBMS
- **Tools:** Git, VS code, PyCharm, Google Cloud Platform, Tableau, PowerBI, WordPress, Azure, Hadoop
- **Libraries:** PyTorch, Plotly, Tensorflow, SciPy, pandas, Scikit-learn, dlib, seaborn, NLTK, XGBoost, NetworkX
- **Platforms:** Microsoft Excel, Google Colab, PySpark, MS Office, Unix/Linux, MySQLWorkbench, R Studio
- **AI & Machine Learning:** AI/ML Experience, Machine Learning Techniques

EDUCATION

Northeastern University | ***Master of Science, Data Analytics Engineering*** **Sep 2022 - Dec 2024**

VJTI | ***B.Tech, Electronics Engineering*** **Aug 2018 - May 2022**

EXPERIENCE

Haas Lab, Mass General Hospital- Harvard Medical School

Research Engineer *Boston, MA* · **Mar 2025 - Present**

- Currently researching stabilizing GNN training on protein interactions through architectural refactoring and hyperparameter tuning, exceeding performance benchmarks by 10% at the current stage under guidence of Dr. Wilhelm Haas.
- Deployed GPU-optimized XGBoost and Neural Networks, enhancing accuracy by 40% over linear models.
- Engineered a graph-based imputation, reducing data sparsity and achieving a 0.07 RMSE, 15% better than NN based methods.

Data Analyst-1 *Boston, MA* · **Jan 2024 - Sep 2024**

- Applied PCA, UMAP, and t-SNE to mass-spec proteomics data using Python (scikit-learn, Scanpy), which improved data quality and revealed distinct breast-cancer subtypes
- Built GNNs on protein-association networks (302 databases) using NetworkX, addressing class imbalance to maximize model performance. Delivered 96% AUC-ROC on node classification models and presented findings to multi-disciplinary research groups.

DataZ

Associate *Remote – Boston, MA* · **Jan 2025 - Mar 2025**

- Built Python-based web scraping pipelines using Selenium, Scrapy, BeautifulSoup and ScraperAPI for specific product websites similar to Alibaba/AliExpress and centralized 30K+ product entries into a MySQL database, enabling structured analytics and automation.
- Delivered actionable insights through Tableau dashboards tracking key engagement metrics, supporting investor updates and product decisions in a lean startup environment.

Saint Louis University

Data Visualization Intern *Remote* · **Oct 2021 - Nov 2021**

- Led Meta Ads analysis as Technical Team Lead, **Architected NLP features** to analyze sentiment trends in user interactions, optimizing Meta Ads performance, and achieving 5% monthly subscriber growth.
- Built interactive data visualizations using Tableau, Seaborn, and Matplotlib that enabled senior managers to identify key trends, leading to more informed business decisions.

Eddy Tools and Tech Solutions

Computer Vision Intern *Remote* · **Jun 2021 - Sep 2021**

- Led an 11-member cross-functional team to deliver a "Live Attention Tracking" POC, meeting all client requirements. Built a real-time attention monitoring tool using facial landmarks using Dlib face recognition pipeline and variance analysis.
- Leveraged Tableau and Power BI to transform complex datasets into actionable business intelligence and statistical visualizations.

AIRPIX

AI, ML AND EDGE COMPUTING INTERN *Mumbai* · **Jun 2020 - Sep 2020**

- Created a large and well-labeled dataset using Python and YOLO object detection algorithm; effective on low-quality CCTV images.
- Collaborated to train a TensorFlow model that identifies personal protective equipment (PPE) in high-risk zones such as industrial sites and mines, improving safety monitoring and reducing missed violations

PROJECTS

Fuel-IT-Right | ***Sundai*** **Nov 2025**

- Live Calorie burn calculator for different intensity level workouts using different sensors on an athlete.
- Encapsulated in a apple watch app and a webpage connected to heart rate and power sensors using Swift Studio.

Satellite Imagery Gap Filling using CNN | ***Wood-well Climate research center*** **Sep 2025**

- Build a CNN Model to predict the gaps in the Landsat dataset for further processing trained on AlphaEarth and modis dataset.
- Ran ETL process using Python on LendingClub data for extraction, cleaned, processed, and fine-tuned the data for future extractions with correlation with the target variable.

KLCE & GIB Regularization - Class Imbalance Resolution **Sep 2023 - Dec 2023**

- Integrated KLCE regularization (ICLR 2024) into GNNs, achieving a 0.9%–29% performance gain over standard log loss functions.
- Applied Graph Information Bottleneck (GIB) techniques to the PubMed dataset, increasing balanced accuracy by 10% through optimized relevance filtering. Architected GNNs using PyTorch Geometric, implementing complex loss functions and conducting rigorous hyperparameter optimization with Optuna.

CredRisk **Sep 2023 - Dec 2023**

- Developed a Credit Risk forecasting model achieving 89% balanced accuracy and a 0.97 specificity score for high-risk loan detection.
- Engineered an ETL pipeline in Python to clean and process LendingClub data, optimizing feature selection via target variable correlation. Synthesized risk insights through interactive Tableau dashboards and Python-based visualization libraries.
- Deployed XGBoost, Random Forest, and Logistic Regression models, utilizing SMOTE oversampling to mitigate class imbalance.

All-in-one Airport Management System **May 2022**

- Using ETL pipelining Created an RDBMS substantially containing the entities in Airport Management system including terminals, pilots, employees, stores, etc. from different data sources. created/ connected the entities with specific access control
- Got insights into the system using CTEs, window functions, and different MySQL Queries.
- Visualized the database using Tableau dashboard and Python Libraries Such as Matplotlib, Plotly, and Seaborn

LEADERSHIP & EXTRACURRICULARS

- Hacker at Sundai MIT- Harvard AI Prototype Launching Club, Kaggle Advanced SQL certificate, Former VP of Relations and Projects, Data Science Hub at Northeastern University, Graduate Certification in Leadership affiliated with Northeastern University.
- Resident Assistant at Northeastern University, Climate Justice Fellow 2023, NU cycling- Road Biking ECCC Racing, WRBB Radio Show Host women in mythology, Fiber artist - Pranpaan, MIT- ESP splash Teacher, Mentor at Wellmet transitional housing.